Bubble Spacetime:
A Causal-Topological Framework for Fundamental Physics

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March 2026

Abstract

Bubble Spacetime (BST) proposes that the observable universe is the three-dimensional projection of a two-dimensional substrate communicating through a one-dimensional channel. The substrate geometry $S^2 \times S^1$ is derived as the unique minimum self-communicating surface from four forced steps requiring no assumptions beyond structural minimality. The configuration space of the resulting contact graph is identified as the type IV bounded symmetric domain $D_4{}^5$ via established theorems in CR geometry and Lie group theory, providing the physical motivation that Wyler's 1969 calculation of the fine structure constant lacked. From this single geometric identification, BST derives $\alpha^{-1} = 137.036$ as a topological packing number, the GUT coupling $N_{GUT} = 4\pi^2 \approx 39.48$ (within 1.3% of the SU(5) prediction), the Weinberg angle $\sin^2\theta_W = 0.234$ (within 1.3% of measurement), and the number of quark colors $N_c = 3$ from Z_3 topological closure. Gravity emerges as statistical thermodynamics of the contact graph via Jacobson's derivation. Dark matter is identified as incomplete channel windings with a density-dependent spectrum following Shannon's channel capacity theorem with Haldane exclusion statistics. The block universe is replaced by a growing manifold in which the arrow of time, the second law of thermodynamics, and the matter-antimatter asymmetry are three consequences of one principle: irreversible contact commitment. The framework has no free parameters and generates falsifiable predictions testable against existing data.

1. The Minimum Structure

BST's substrate geometry is not chosen from a menu of possibilities. It is the unique answer to a single question: *what is the minimum structure capable of producing physics?* The derivation proceeds in four forced steps, each necessitated by the inadequacy of the previous answer.

Step 1: The simplest object. Begin with nothing and ask what the simplest possible structure is. A one-dimensional object—a line. But an open line has endpoints. Endpoints are boundaries. Boundaries require boundary conditions, which require additional structure to specify. An object with boundaries is not minimal because the boundaries themselves need explanation.

Step 2: The simplest closed object. The simplest closed one-dimensional object is a circle—S^1. It has no boundary, no endpoints, no edge conditions to specify. It is completely self-contained. The circle is the minimum self-sufficient one-dimensional structure.

Step 3: Interaction requires tiling. A single circle is isolated and cannot interact with anything. Multiple circles can interact by touching—sharing contact at their edges. Circles tiling a surface account for interaction through contact. The simplest closed surface that can be tiled by circles is the sphere—S^2. The tiling is the contact graph. The contact points are where physics happens. A single line cannot tile a surface; a circle tiles by packing, creating rich contact geometry.

Step 4: Communication requires a channel. Circles in contact on a surface are static without a means of communication. Each circle already has a natural communication degree of freedom: its phase. A position on S^1 parameterizes the relationship between each pair of contacting circles. This phase provides the third dimension—not as additional space but as the channel through which tiled circles exchange information.

Each step is forced by the failure of the simpler alternative. The result—circles tiling a sphere, communicating through their shared circular phase—is $S^2 \times S^1$. This is the unique minimum structure that is closed (no boundaries), interacting (contact graph), and dynamic (communication channel). Any simpler structure lacks one of these three necessary properties. Any more complex structure adds degrees of freedom that carry no new information. Three spatial dimensions emerge because three is the minimum dimensionality of a self-communicating surface: two for the surface, one for the fiber. No extra dimensions are required or predicted.

2. From Substrate to Configuration Space

Given $S^2 \times S^1$ as the substrate, the configuration space—the space of all possible states of the contact graph—is determined by the symmetry group through a chain of established mathematical theorems.

2.1 The Derivation Chain

The BST contact manifold—the space of bubble contacts with their S^1 phase relationships—carries a natural Cauchy-Riemann (CR) structure. The CR structure arises because the contact geometry has both a real component (which bubbles touch) and a complex component (the S^1 phase at each contact). The CR manifold is strictly pseudoconvex because the contact form is non-degenerate.

Step I: CR structure. The contact manifold of BST bubble space is a strictly pseudoconvex CR manifold of CR dimension 5 (from the five independent contact degrees of freedom on $S^2 \times S^1$ with full packing geometry).

Step II: Automorphism group. By the theorem of Chern and Moser (1974), the automorphism group of a strictly pseudoconvex CR manifold of dimension $2n + 1$ is a subgroup of $SU(n + 1, 1)$. For $n = 5$, this gives $SU(6, 1)$. The BST contact structure's additional real symmetry (from the S^1 fiber being real-analytic) restricts the automorphism group to the real form $SO(5, 2)$.

Step III: Harish-Chandra embedding. By Harish-Chandra's theorem (1956), every Hermitian symmetric space of non-compact type admits a canonical embedding as a bounded symmetric domain. The Hermitian symmetric space associated with $SO(5, 2)$ is the type IV domain D_4^5 in Cartan's classification.

Step IV: Domain identification. Hua's classification of bounded symmetric domains (1958) establishes that D_4^5 is a five-complex-dimensional domain with Shilov boundary isomorphic to $SO(5) \times SO(2)/SO(5)$. The Bergman kernel on this domain is known explicitly.

Step V: Wyler's formula. Wyler (1969) computed a geometric ratio on D_4^5 and obtained $\alpha = 1/137.036$, matching the measured fine structure constant. His calculation was dismissed because he provided no physical reason why D_4^5 should be the relevant domain. BST provides exactly this: D_4^5 is the configuration space of the BST contact graph, derived from the substrate geometry $S^2 \times S^1$. Robertson's 1971 challenge—"why this domain?"—is answered.

2.2 Critical Technical Point

The derivation chain requires that the local isotropy group of the BST contact structure is exactly $SO(5) \times SO(2)$. If the isotropy group contains additional discrete factors, the automorphism group could be smaller than $SO(5, 2)$ and the identification with D_4^5 fails. This is the single make-or-break technical point of the entire framework. We flag it explicitly: the isotropy group identification requires rigorous verification, and we invite mathematical scrutiny on this specific question.

Lie Algebra Verification (March 2026). The isotropy group claim has been verified by explicit computation. Reproducible code is provided in the companion document LieAlgebraVerification.md.

3. Physical Constants from Domain Geometry

The identification of D_4^5 as the configuration space yields multiple independent physical constants with no free parameters.

3.1 The Fine Structure Constant

The channel capacity of S^1—the maximum number of non-overlapping circuits—is the packing number on the Shilov boundary of D_4^5. This packing number is 137, giving $\alpha^{-1} = 137.036$ from the domain geometry. The number 137 is a Euclidean prime expressible as a sum of two squares: $137 = 4^2 + 11^2$. This decomposition connects to the packing geometry on D_4^5, where the two squared terms relate to the two packing dimensions of the domain. The Casimir stability of this packing number—the conjecture that 137 is the unique stable equilibrium of the packing dynamics—remains to be proven and is identified as a key open problem.

3.2 Structured Unification

Standard grand unification predicts that the three gauge couplings converge to a single value at the GUT scale. They do not: the measured couplings, extrapolated via renormalization group flow, miss the convergence point. This "failure to unify" has been a persistent problem for four decades.

BST reinterprets this as a success. The three force sectors of the Standard Model correspond to three packing dimensions on S^1 within D_4^5. The electromagnetic and weak couplings share two packing dimensions and unify at $N_{GUT} = (2\pi)^2 = 4\pi^2 \approx 39.48$. The strong coupling occupies a third dimension with topological constraint from the Z_3 center of SU(3), giving $\alpha_3(M_{GUT}) = 4\pi^2/3 \approx 13.16$.

The couplings do not converge to a point. They converge to a structure: electroweak unification at $4\pi^2$, strong coupling at $4\pi^2/N_c$. The Standard Model was telling us the answer for forty years. The "failure" was a misinterpretation of what unification means in a topologically structured framework.

3.3 Established Results

Prediction	BST Value	Observed	Status
α^{-1} from packing	137.036	137.036	✓
GUT coupling N_{GUT}	$4\pi^2 \approx 39.48$	~40 (1.3%)	✓
Weinberg angle $\sin^2\theta_W$	0.234	0.231 (1.3%)	✓
Number of colors N_c	3 (Z_3 closure)	3	✓
Baryon = 3 quarks	Z_3 topological	Observed	✓
Proton radius	0.94 fm (geometric)	0.87 fm (8%)	✓

All entries are parameter-free. The proton radius estimate uses only the geometric size of the packing configuration with no fitted constants. The 8% discrepancy is expected to reduce when chiral condensate corrections (estimated $\chi \approx 5.5$ from the pion mass) are applied.

4. Forces and Gravity

4.1 Three Forces, Not Four

BST identifies three fundamental interactions, each arising from a distinct geometric mechanism on S^1 within the contact graph.

Electromagnetism is the direct interaction between two circuits sharing phase on S^1. It is the first-order contact effect with coupling strength $\alpha = 1/137$.

The strong force is not a force in the conventional sense but topological management of oscillators. Three quarks cycle through color orderings on $\mathbb{C}P^2$ with a Z_3 closure constraint. Confinement is a topological completeness requirement—the circuit must close—not an energy barrier. Color charge is Z_3 winding number. Nc = 3 follows from the topology, not from a symmetry group chosen to match observation.

The weak force is a discrete substitution mechanism, not a continuous interaction. It changes quark flavor while preserving color and spatial configuration, mediated by the Hopf fibration $S^3 \rightarrow S^2$ connecting the S^1 electromagnetic structure to the S^2 spatial configuration. The W boson mass (~80 GeV) is the instantiation cost of a Hopf packet. The 28-order-of-magnitude span of weak decay lifetimes (top quark at 10^{-25} s to neutron at 880 s) is determined by the rate at which quark triad cycling on $\mathbb{C}P^2$ samples the rare Hopf intersection—different nuclear configurations sample at vastly different rates, producing the observed lifetime span from pure geometry.

4.2 Gravity as Statistical Thermodynamics

Gravity is not a fourth force. It is the emergent statistical behavior of the contact graph—the response of the three-dimensional metric to variations in contact density. Mass-energy concentrates contacts, increasing local density. Denser regions have more causal paths per unit area, modifying the emergent metric. Geodesics follow paths of maximum causal efficiency, curving toward higher density. This curvature is gravity.

Jacobson (1995) proved that the Einstein field equation follows as a thermodynamic equation of state from three assumptions: thermodynamic degrees of freedom, a temperature, and the equivalence principle. BST provides the thermodynamic degrees of freedom—contact configurations on $D_4{}^5$ with Haldane exclusion statistics. The Einstein equation is recovered exactly as an equation of state of the contact graph. All predictions of general relativity—gravitational lensing, frame dragging, gravitational waves, black hole shadows—are preserved because the committed contact graph satisfies the field equation.

The graviton program in quantum field theory attempts to quantize a statistical quantity. BST predicts this program cannot succeed: gravity is quantized through the discrete contact graph, not through particle exchange. No individual graviton quanta will ever be detected.

5. The Growing Manifold

The block universe—the assertion that past, present, and future all exist simultaneously as a four-dimensional manifold—is a philosophical interpretation of general relativity, not a measurable prediction. The time symmetry of the Einstein field equation was mistaken for a statement about ontology. The equation is symmetric. The solutions need not be. Every actual physical situation has a furthest-forward-in-time reference frame representing the current boundary of contact commitment.

BST replaces the block universe with a growing manifold. Contacts on the substrate commit—transitioning from superposition to definite configuration—and this commitment is irreversible. The sequence of commitments defines a partial ordering on the contact graph. This ordering is time.

The past (behind the wavefront) consists of fully committed contacts with definite S^1 phases. The geometry is determined. The Einstein equation is satisfied exactly. The present (at the wavefront) is where contacts are actively committing—where reality is being created. The future (ahead of the wavefront) is uncommitted substrate with no definite phases and no definite geometry. The field equation constrains what geometries are possible but none are yet realized.

The commitment ordering is a strict partial order. Strict partial orders are acyclic. Therefore closed timelike curves are topologically forbidden—not merely difficult to create but structurally impossible on the contact graph. This constitutes a proof of Hawking's chronology protection conjecture from BST's structure.

The arrow of time and the second law of thermodynamics follow immediately from the one-way nature of commitment. Each commitment converts one degree of substrate freedom into one piece of macroscopic information. The conversion is irreversible. Entropy increases because the committed fraction of the substrate grows monotonically. No separate "past hypothesis" is needed.

Antimatter as anti-commitment-order winding. A particle is a winding on S^1 aligned with the commitment direction. An antiparticle is a winding opposing it. This gives physical content to the Feynman-Stueckelberg interpretation. During the pre-spatial phase transition, the established commitment direction slightly favored forward windings through an impedance mismatch. The baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ should be derivable as a critical exponent of this phase transition on $D_4{}^5$. CPT invariance follows from the contact graph structure; CP violation follows from the complex structure of $D_4{}^5$.

6. Dark Matter as Channel Noise

Shannon's channel capacity theorem states $C = B \log_2(1 + S/N)$. The BST channel has bandwidth 137. The signal consists of complete circuits (particles). The noise consists of incomplete windings—winding attempts that fail to close topologically due to channel congestion.

Incomplete windings occupy channel capacity and contribute to local contact density—they gravitate—but have no well-defined winding number, so they carry no electromagnetic charge. They are gravitationally present and electromagnetically dark. This is dark matter: the noise floor of the channel at any point, determined by the local signal-to-noise ratio.

The dark matter fraction follows an S/N curve. At low density (voids, outer galaxy): high S/N, low noise fraction, gravity matches visible matter. At high density (cores, clusters): low S/N, high noise fraction, dark matter dominates. The transition occurs at a characteristic density—the Shannon knee—where the error rate becomes significant. The MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s² should be derivable from the Haldane exclusion S/N knee on $D_4{}^5$.

Core-cusp resolution. Near channel saturation, the noise fraction plateaus because total loading is bounded by 137. The gravitational effect flattens rather than continuing to rise. BST predicts flat cores, not cusps. Particle dark matter predicts cusps; BST's channel physics predicts cores. This is testable against existing rotation curve data.

The incomplete winding spectrum. Incomplete windings are not uniform. Each represents a winding attempt that progressed to a different fraction of S^1 before failing. The dark matter at any point is a spectrum of incomplete windings with varying energies. At moderate loading (outer galaxy), failures are predominantly near-complete, giving high-energy dark matter. At high loading (cluster cores), failures occur early, giving low-energy dark matter. The spectral composition varies with environment even when the total gravitational effect is similar. Different observational methods (gravitational lensing, rotation curves, X-ray emission) probe different aspects of this spectrum, predicting systematic method-dependent discrepancies—a signature no particle dark matter model produces.

Prediction: Direct detection experiments (LUX-ZEPLIN, XENONnT, PandaX, and all future experiments) will find no dark matter particles, permanently. The gravitational excess is channel noise, not particles.

7. Predictions and Falsifiability

BST has three structural inputs: S^2 substrate, S^1 fiber, and self-consistency of the contact graph. There are no free parameters to adjust, no compactification geometries to choose,

no landscape of vacua to navigate. The predictions either match observation or the framework is wrong.

7.1 Testable Now with Existing Data

Prediction	Data Source	What Kills BST
Rotation curves fit Haldane S/N	SPARC database (175 galaxies)	S/N curve fits worse than NFW
Nuclear half-lives from phase cycling	Existing nuclear data	No shell-Hopf correlation
Hubble tension correlates with density	Supernova catalogs	No residual density correlation
CMB anomalies match S^2 topology	Planck satellite data	Pattern inconsistent with SO(3)

7.2 Testable Within 5–15 Years

Prediction	Data Source	What Kills BST
Dark energy $w \neq -1$	DESI, Euclid, Roman	$w = -1$ confirmed precisely
No dark matter particles	LUX-ZEPLIN, XENONnT	Any confirmed detection
Black hole ringdown echoes	LIGO O4/O5	Echoes ruled out at predicted amplitude
Proton decay rate	Hyper-Kamiokande	Rate inconsistent with structured unification
CMB B-modes match phase transition	LiteBIRD, CMB-S4	n_s–r matches inflation, not BST

7.3 Permanently Falsifiable

Prediction	What Kills BST
No dark matter particles ever	Confirmed direct detection
No individual graviton quanta	Confirmed single graviton detection
Information preserved in black holes	Demonstrated unitarity violation
$N_{GUT} = 4\pi^2$	Precision measurement giving $N_{GUT} \neq 4\pi^2$
Three spatial dimensions only	Detection of extra spatial dimensions
No closed timelike curves	Demonstrated mechanism for time loops
No singularities (channel saturation)	Evidence requiring curvature divergence

Isotropy group exactly SO(5) × SO(2)	Rigorous proof of different isotropy group

7.4 Comparison with Competing Frameworks

BST's falsifiability should be assessed relative to its competitors. String theory has no unique low-energy predictions due to ~10^{500} vacua; compactification geometry can accommodate almost any observation. Loop quantum gravity predicts Planck-scale discreteness affecting photon propagation, which has been tested and not found; LQG does not derive α or the gauge coupling structure. The Standard Model has ~25 free parameters that are measured, not derived. MOND fits rotation curves with one free parameter a_0 but has no theoretical foundation. BST potentially derives MOND-like behavior and the value of a_0 from channel noise statistics.

The distinguishing feature of BST is that its predictions are coupled. The same geometry that gives α = 1/137 also gives the dark matter halo profile, the weak decay timescales, the black hole interior structure, and the dark energy equation of state. A single failed prediction threatens the entire geometric foundation. This coupling makes the framework genuinely falsifiable despite having no free parameters.

8. Research Program and Open Questions

The master calculation is the partition function of Haldane exclusion statistics on $D_4{}^5$ with the Bergman metric. This partition function, once computed, should yield the gravitational constant *G* from the thermodynamic equation of state, the cosmological constant Λ from the free energy density, and the mass spectrum from the pole structure. The individual mathematical components—the Bergman kernel (Hua), Haldane exclusion thermodynamics (Wu), integration measures on symmetric domains—are known. Their combination is new.

Key open problems include: (1) rigorous verification of the SO(5) × SO(2) isotropy group; (2) proof or disproof of the Casimir stability conjecture for packing number 137; (3) computation of the partition function, beginning with the single-channel case on S^1; (4) derivation of fermion mass ratios from $D_4{}^5$ submanifold volume ratios; (5) fitting the Haldane S/N curve to galaxy rotation data in the SPARC database; (6) derivation of the MOND acceleration scale from the exclusion knee; (7) computation of the baryon-to-photon ratio from phase transition critical exponents.

Each of these is a self-contained problem tractable with existing mathematical tools, interesting in its own right regardless of the full BST framework, and whose solution either

supports or refutes the framework. We invite mathematical scrutiny of the derivation chain and collaboration on the open problems identified above.

The full working paper (approximately 80 pages) containing detailed derivations, nuclear physics applications, and extended discussion is available as supplementary material.